

Decision memo: claims intake process redesign

Prepared as a data analytics / business analysis portfolio project · all data synthetic

Situation

Insurance claims intake averages 12.48 days end-to-end, with wide variance by claim type. A quantified process-mining analysis of a 5,000-claim synthetic event log (57,781 events) was conducted to identify where the process loses time and what specifically should change.

What was analyzed

Cycle time by activity, rework loop frequency, handoff delay isolated from processing time, first-contact resolution (FCR), and cycle time by channel and claim type — computed with DuckDB directly over the event log. Full methodology, code, and analysis are in the project repo.

What was found

- **Injury claims are the weak point on every metric** — 59.8% FCR, 38.4% documentation rework rate, 20.63-day average cycle time — driven by a 7.44-day average wait on third-party medical records, the single largest cycle-time driver anywhere in the process.
- **Documentation rework** (16.4% of all claims, 5.52-day average wait once triggered) is the second-largest driver, concentrated in Property and Injury claims.
- **Manager review is real but narrower than assumed going in.** Only 9.6% of claims trigger it, and the average wait (1.28 days) is misleading — the median is 0 days. The actual problem is a 4.91-day p90 tail from periodic congestion, not a permanent capacity shortfall.
- **Channel adds a smaller, consistent effect** — self-service (Web/Mobile) submissions run 0.7–1.3 days slower than staff-assisted (Phone/Agent) ones, in every claim type.

What is recommended

- Same-day claim registration on every intake channel
- A document completeness check at intake, catching gaps before they reach an adjuster
- Medical records requested in parallel with adjuster review for Injury claims, not after it
- Widened, business-configurable fast-track eligibility thresholds
- A small overflow reviewer pool, triggered only during manager-queue congestion — not a permanent fifth manager

What it is worth

Each change was implemented as actual process logic in a second simulation (same resource-queue mechanics as the as-is model) rather than estimated by hand:

Metric	As-is	To-be	Change
Avg cycle time (blended)	12.48 days	9.03 days	-27.6%
Injury cycle time	20.63 days	13.92 days	-32.5%
First-contact resolution	82.8%	94.3%	+11.5 pts
Documentation rework rate	16.4%	~5.0%	-11.4 pts
Manager-review p90 wait	5.12 days	0.0 days	overflow absorbs tail

All figures are from a synthetic event log and a process simulation, not a live system. Impact figures are projections and should be re-validated against production data if implemented. Full methodology, code, BRD, user stories, requirements traceability matrix, and UAT test plan are in the accompanying repository.